

MEDICAL MONITORING AND ADVANCED SENSOR DATA PROCESSING – TASK 02: TIME FREQUENCY ANALYSIS

1 General Information

The practical experiments should be done in groups of up to three people. Each group must submit one solution via Digicampus. Of course, you can discuss the tasks between groups, but forwarding/copying other students' source code is forbidden.

Prepare the task until the deadline provided below/in Digicampus.

On Digicampus, a zip archive contains a template for you to prepare your code for submission. Please complete your code within this template. The **README.md** file provides information on how to set up your Python environment for running your code. A Jupyter notebook example is also provided. You can adapt the content of this Jupyter notebook for preparing your solution.

Before uploading your solution, please make sure that the following requirements are met:

- Prepare your result documentation as Jupyter Notebook
- **Print a PDF version of your Jupyter notebook AND add it to the .zip archive you upload on Digicampus.**
- All source code must be available and executable for us (this implies that you specify the required packages/libraries. In addition, you have to provide necessary data (ideally data should be in a subfolder, and you use relative paths) and/or give advice if paths have to be adjusted; in order to do so, you should provide a reasonable README file)
- **GREEN** sections in the chapter on tasks (see section **Error! Reference source not found.**) require consideration and content in your Jupyter notebook. Please include a numbering according to the questions' number Q<xy>
- List your sources (if applicable)

Submission deadline is **Thursday, 12.06.2025, 08:00**; submission is done via **Digicampus**.

Please ask in case of any doubts regarding the contents or the formal procedure.

2 Elements to be delivered

1. Jupyter Notebook: the students must deliver the code (as .ipynb file) they developed to visualize, analyse and plot the data according to the template provided.
A pdf version of the Jupyter notebook with the results of the notebook's run should be provided as a backup in case the notebook fails to be run again.

NOTE:

A requirements.txt file is provided on digicampus. Students HAVE TO USE ONLY the libraries that are provided within this file.

2. The students' recordings that are necessary to execute their Jupyter notebooks.
3. Report: the students must deliver a report (as .pdf file) using the ReportTemplate.docx found in Digicampus. The report must content the following:
Title Page
Title: Clear and specific
Course name / lab name
Students names
Date
Answer to the questions (with the appropriate question numbering)
4. When a question asks to discuss a topic, the word limit is an upper limit.

3 Remarks on the Hardware and software

A documentation to perform the recordings based on the smartwatch and Biopac platform is provided [1]. Executables to perform recordings via the ADI Study watch are provided in a Flashdrive [2]. Those executables should be copied to the computer which will record the data.

3.1 Biopac

One can export the acquired data in a format that is suitable for postprocessing.

3.2 ADI Study Smartwatch

The communication between the smartwatch and the computer is established via a Dongle connected to a plugged USB port. One needs to check the port address (COM) using the device manager.

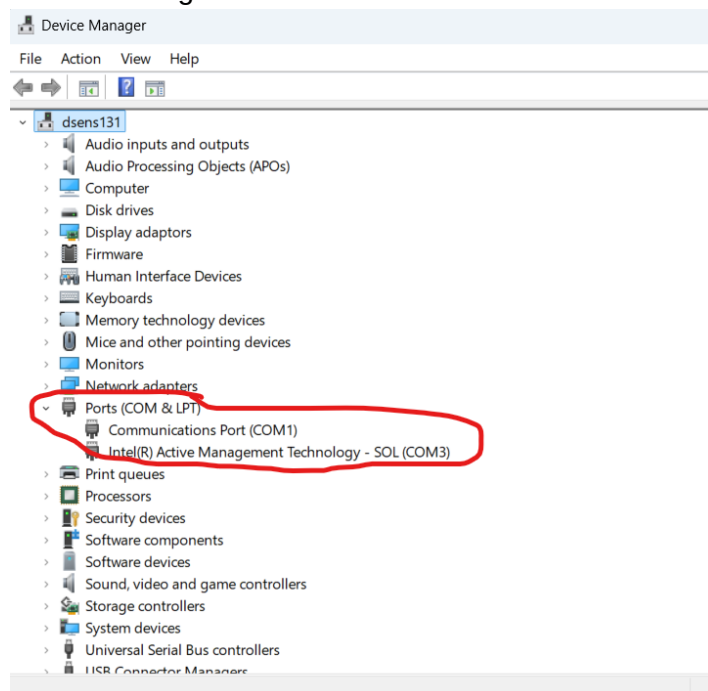


Figure 1 Device manager (on Windows).

The recordings are stored on the computer in .csv files.

The smartwatch can provide an electrocardiographic signal (ECG), a photoplethysmographic signal (PPG), and accelerometer signals (along the 3 directions).

4 Remarks on reports and data collection

Following some remarks on the report and data collection:

- Reports for data gathering should be detailed and complete. Students will use the data they collect to work on it by themselves.
- The report must assure the reproducibility of the experiment and data gathering, bearing in mind the amount and quality of data that is intended to be collected.
- If you cite from the literature, use the APA citation style
- For plots in the time domain, signals must be plotted with x-axis in time, in seconds.
- All plots must have axes labelled in a meaningful way.
- All plots or images must have caption written in a meaningful way.
- In the caption, do not forget to refer to the question number **Q...** the plot is built for.
- If a question asks you to make a plot, include the plots with appropriate reference to the question in the caption, and write a reference to the plot in the report.

For example :

Qxx. Area under the curve obtained on the test set by several classifiers is displayed on Figure 2.

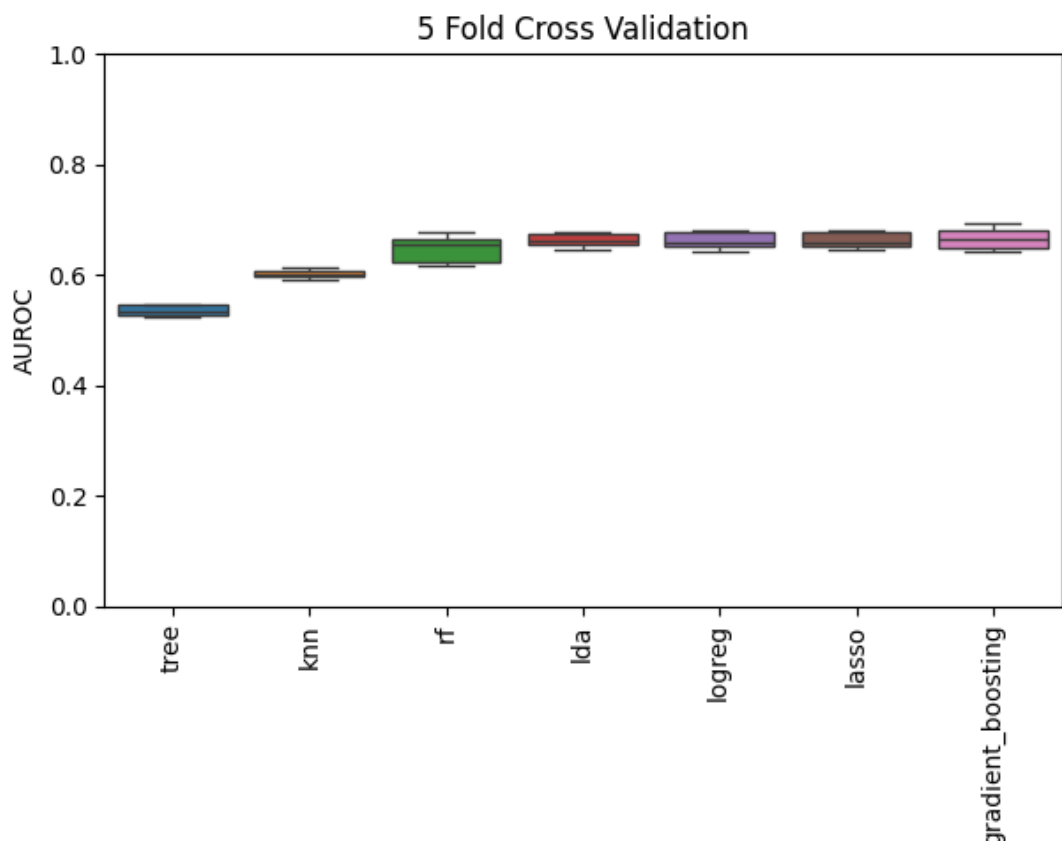


Figure 2 Area under the curve of several classifiers on the test set (**Qxx**).

- **ATTENTION:** Have a look at the general document on our expectation and common mistakes, which is available in Digicampus

5 Tasks

5.1 Recordings

For this exercise you need to perform the following recordings. For all recordings you use Biopac and the smartwatch simultaneously. PPG signals (wrist and finger) are to be recorded at the same hand.

- Clean signals
 - ECG and PPG signals 3 minutes recording during rest
- Noisy signals
 - ECG and PPG signals 3 minutes use the power supply (Makita) for introducing noise
 - ECG and PPG signals 3 minutes move your hand with PPG sensor and smartwatch

5.2 Time-domain analysis

Q1. Consider a one-minute segment of two ECG signals, one with noise and the other without, and for them (0.5 Points):

a. Plot both signals and compute the following descriptive measures:

- Mean
- Median
- Variance
- Standard deviation (Std Dev)
- Skewness
- Root Mean Square value (RMS)
- Zero-Crossing Rate (ZCR)
- Energy
- Power

Discuss, in one paragraph, about what you observe from the descriptive measures computed from the signals with and without noise.

b. Consider the following two windowing strategies: fixed with constant duration of ten seconds without overlap, and with overlap of 50% (five seconds). For each window, compute the descriptive measures listed in the previous item. These procedures must be applied to the two signals considered in the previous item. Organize the results in tables, as the following

ECG segme nt	Mean	Media n	Varian ce	Std Dev	Skewne ss	RMS	ZCR	Energy	Power
1									
2									
3									

4									
5									
6									

Discuss, in one paragraph (max. 200 words), about what you observe when comparing the two windowing strategies implemented.

5.3 Frequency-domain analysis

Q2. Consider a ten-second segment of two ECG signals, one with powerline noise and the other without, and for them (1 Point):

- a. Compute the DFT. Plot the signals in the time domain and the absolute value of the DFT.

Discuss, in one paragraph (max. 150 words), the spectrums computed from both signals. What information can you get from them?

- b. Compute the Power Spectral Density (PSD) from the signals analysed in the previous item. For this, consider the periodogram and the Welch's method. From the PSD calculated with the Welch's method, compute the Spectral Centroid and Spectral Entropy. Compare two ECG, one from the chest (Biopac recording) and one from the wrist (smartwatch). What happens to the spectra if noise is introduced?

Discuss, in one paragraph (max. 150 words), the PSDs computed from both signals. What information can you get from them? What is the difference in comparison with the spectrums from the previous item?

Q3. Consider two finger PPG signals (1 at rest, 1 with motion noise) and do the same as question Q2 for those signals. (1 point)

Q4. In the provided notebook, 3 functions are provided (0.5 points):

generate_composite_signal generates a signal summing sinusoids at specified frequencies.

generate_non_stationary_signals generates a composite signal concatenating sinusoids of different frequencies over time, and its time-reversed version.

Compute the spectrum for all the generated signals. Plot the signals and the power spectra.

Discuss, in one paragraph (max. 150 words), the spectra computed from the signals. What can you observe on them? What might you conclude?

5.4 Time-frequency-domain analysis

5.4.1 From Short time Fourier Transform To Discrete Wavelet Transform

Q5. Considering the signals generated in Q4, compute the Short Time Fourier Transform (STFT) ONLY for the non-stationary signal (variable `signal_non_st`), and its reversed version (variable `reversed_signal_non_st`). Plot the signals and the STFT (0.5 points).

Discuss, in one paragraph (max. 150 words), the STFT computed from the signals. What can you observe on them? What might you conclude? Compare the information provided by the STFT in comparison with the Fourier transform and the PSD.

Q6. (0.5 points)

- a. Compute the Short Time Fourier Transform (STFT) of a ten-second segment of two ECG signals, one with noise and the other without. Plot the signals and the STFT.
- b. Compute the STFT of a ten-second segment of two PPG signals (finger and wrist) with motion noise. Plot the signals and the STFT.

Discuss, in one paragraph (max. 150 words), the STFT computed from the ECG and PPG signals. What can you observe on them? What might you conclude?

Q7. (0.5 points)

- a. Compute the Continuous wavelet transform (CWT) of a ten-second segment of two ECG signals, one with noise and the other without. Plot the signals and the CWT. Experiment with different mother wavelets.
- b. Compute the CWT of ten-second segment of two PPG signals. Plot the signals and the CWT. Experiment with different mother wavelets.

Discuss, in one paragraph (max. 150 words), the CWT computed from the ECG and PPG signals. What can you observe on them? What might you conclude? What changes do you observe when changing the mother wavelet?

Q8. (0.5 points)

- a. Compute the discrete wavelet transform (DWT) of a ten-second segment of two ECG signals, one with noise and the other without. Plot the signals and the detail and approximation coefficients. Experiment with different levels of decomposition and mother wavelets.
- b. Compute the DWT of a ten-second segment of two PPG signals. Plot the signals and the detail and approximation coefficients. Experiment with different levels of decomposition and mother wavelets.

Discuss, in one paragraph (max 150 words) the DWT computed from the ECG and PPG signals. What can you observe on them? What might you conclude? What changes do you observe when changing the levels of decomposition and mother wavelet?

5.4.2 Denoising Using Discrete Wavelet Transform

Q9. Applying DWT in ECG signal processing (0.5 point).

Baseline wander correction

- a. Take a twenty-second segment of an ECG signal. Determine the maximum useful level of decomposition for that signal. Bear this in mind for the following steps.
- b. Perform DWT decomposition. Plot the signal and decomposition coefficients.

- c. Store baseline approximation coefficients in a variable and make detail coefficients equal to zero. Reconstruct the baseline. Plot the reconstructed baseline.
- d. Remove baseline wander by zeroing approximation coefficients and keeping detail coefficients. Store those coefficients in a variable. Reconstruct the signal without the baseline. Plot the original and reconstructed signal without baseline.
- e. Experiment with the mother wavelets 'db4', 'sym4', 'bior3.9', and the levels of decomposition 6, 8, 10. Plot the original signal and the signal without baseline for each case.

Discuss, in one paragraph (max. 200 words), about the followed procedure and the results you got. What might you conclude from them? What changes do you observe when changing the levels of decomposition and mother wavelet?

Q10. Denoising using the DWT (0.5 points)

- a. Take a twenty-second segment of an ECG signal. Determine the maximum useful level of decomposition for that signal. Bear this in mind for the following steps.
- b. Perform DWT decomposition. Plot the signal and decomposition coefficients (samples in horizontal axis).
- c. Filtering process is performed by zeroing detail coefficients (that usually contain the noise) and afterwards reconstructing the signal.

Experiment by zeroing different detail coefficients D1, D2, D3. Reconstruct the signal. Plot the original and filtered signal.

- d. Experiment with the mother wavelets 'db4', 'sym4', 'bior3.9', and the levels of decomposition 6, 8, 10. Plot the original signal and the signal without noise for each case.

Discuss, in one paragraph (max. 200 words), about the followed procedure and the results you got. What might you conclude from them? What changes do you observe when changing the levels of decomposition and mother wavelet?

6 Material (and recommended readings)

- [1] Data Recordings- Overview.pdf
- [2] Flashdrive with the executables for the smartwatch
- [3] Numpy and Scipy Documentation — Numpy and Scipy Documentation.
docs.scipy.org/doc. Accessed 21 May 2025.
- [4] NumPy Documentation. numpy.org/doc. Accessed 21 May 2025.
- [5] PyWavelets - Wavelet Transforms in Python — PyWavelets Documentation.
pywavelets.readthedocs.io/en/latest. Accessed 21 May 2025.
- [6] Matplotlib — Visualization With Python. matplotlib.org. Accessed 21 May 2025.